

Macroeconomic Theory and Policy: Lecture W3b

What are You Going to Learn

- Proximate the output trends/potential output
- Calculate moving average
- Introduce Hamilton Filter

Moving Averages (MA)

- Calculating the average of fixed number of the most recent observations in a rolling window
 - ▶ e.g., MA of 3 quarters for 2001Q1 can be the average of 2000Q3, 2000Q4 and 2001Q1
 - ▶ Then, we move to 2001Q2, the MA includes 2000Q4, 2001Q1 and 2001Q2
 - ▶ Continue like this for the whole data series
- The goal is to smooths out short-term fluctuations
- Highlights longer-term trends or cycles

Trailing and Centered Moving Averages

- *Trailing MA (Causal MA)*: Uses only past data points up to the current time.
 - ▶ Example: A 5-period trailing MA at time t is:

$$MA_t = \frac{1}{5} \sum_{i=t-4}^t X_i$$

- *Centered MA (Symmetric MA)*: Averages data from both past and future points.
 - ▶ Example: A 5-period centered MA at time t is:

$$MA_t = \frac{1}{5} \sum_{i=t-2}^{t+2} X_i$$

Leading Moving Average

- *Leading MA (Forward MA)*: Uses only future data points, which is uncommon in real-world applications.
 - ▶ Example: A 5-period leading MA at time t is:

$$MA_t = \frac{1}{5} \sum_{i=t}^{t+4} X_i$$

- Most macroeconomics time series data would use Trailing MA, because economic agent at a specific time would not have any exact information about the future

Simple Moving Average (SMA)

- Basic form of moving average
- This is what we have been showing so far, for Trailing MA,

$$SMA_t = \frac{1}{N} \sum_{i=t-N+1}^t X_i$$

- Equal weights for all observations (the weight is $1/N$)

Simple Moving Average (SMA): Example

- Suppose we have stock prices for the past 5 days:

$$X_1 = 100, \quad X_2 = 102, \quad X_3 = 104, \quad X_4 = 103, \quad X_5 = 105$$

- The 3-day SMA at time 3 is:

$$SMA_3 = \frac{100 + 102 + 104}{3} = 102$$

- The 3-day SMA at time 4 is:

$$SMA_4 = \frac{102 + 104 + 103}{3} = 103$$

- The 3-day SMA at time 5 is:

$$SMA_5 = \frac{104 + 103 + 105}{3} = 104$$

Weighted Moving Average (WMA)

- Assigns different weights to observations (e.g., giving more importance to recent data)
- Unlike the Simple Moving Average (SMA), where all points are equally weighted, WMA emphasizes more recent values
- The formula for WMA over a window size N is:

$$WMA_t = \frac{\sum_{i=t-N+1}^t w_{i-t} X_i}{\sum_{i=t-N+1}^t w_{i-t}}$$

where w_{i-t} represents the assigned weight for each observation (e.g., w_{-1} is the weight on the X that is one period lag, w_{-2} is the weight on the variable that is 2 period lag)

Drawbacks of Moving Averages

- *End-Point Problems:*

- ▶ At the start and end of the data series, fewer past or future values are available, leading to missing or biased estimates

- *Lag in Identifying Turning Points:*

- ▶ Since moving averages rely on past data, they respond slowly to sudden changes, delaying trend identification

- *May Miss Structural Breaks:*

- ▶ Moving averages smooth out fluctuations, which may cause important shifts, such as economic crises or policy changes, to be overlooked

Drawbacks of Moving Averages

- *Arbitrary Selection of Weights:*
 - ▶ In weighted moving averages, the choice of weights is subjective and may not always reflect true data dynamics
- *Sensitivity to Window Length Selection:*
 - ▶ A short window reacts quickly but may capture too much noise, while a long window smooths better but introduces more lag

Introduction to Hamilton Filter

- Developed by James Hamilton from UCSD (Hamilton, 2018)
- The original paper shows it is an alternative to another popular smoothing method Hodrick-Prescott filter (we won't go into detail on the HP filter)
- Based on regression framework which provide more basic statistical properties
- Addresses traditional filter limitations

Hamilton Filter: Basic Concept

- Projects future values based on recent past
- Focuses on h -period ahead prediction
- Uses linear regression methodology (more specifically, time series econometric)

Hamilton Filter Mathematics

- *Basic Equation:*

$$y_{t+h} = \beta_0 + \beta_1 y_t + \beta_2 y_{t-1} + \beta_3 y_{t-2} + \beta_4 y_{t-3} + v_{t+h}$$

The equation uses lagged values of y to predict y_{t+h} , where v_{t+h} captures the cycle component (deviations from trend)

- *Cycle Component:*

$$v_{t+h} = y_{t+h} - \text{predicted value}$$

This isolates the cyclical component by subtracting the predicted trend value from the observed value

- *Preserves Time Series Properties:* Unlike other filters, the Hamilton Filter does not require detrending or smoothing assumptions, retaining the original properties of the time series

Why 4 Lags and $h = 8$?

- *Choice of 4 Lags:*

- ▶ Quarterly data often exhibit seasonal patterns
- ▶ Using 4 lags captures one full year of quarterly dynamics, accounting for both seasonal effects
- ▶ Ensure the residual term is stationary – i.e., bound up and down around zero (you will learn more in time series econometric)

- *Choice of $h = 8$:*

- ▶ $h = 8$ corresponds to a 2-year horizon for quarterly data, which is long enough to separate trends from short-term cycles
- ▶ This choice avoids blending cyclical movements with longer-term trends, ensuring clear distinction between the two

Advantages of the Hamilton Filter

- *Theoretically Well-Founded:*

- ▶ Based on standard regression principles, it uses lagged observations to model trends and cycles.
- ▶ This avoids arbitrary smoothing parameters and is grounded in economic theory.
- ▶ Sometime refer as

- *More Reliable in Real-Time:*

- ▶ Because it uses only past data (causal filtering), it provides realistic estimates for real-time decision-making.
- ▶ No need for future data, making it ideal for forecasting and policy applications.

Advantages of the Hamilton Filter

- *Preserves Basic Time Series Properties:*
 - ▶ Does not distort key properties like stationarity, seasonality, or autocorrelation.
 - ▶ Ensures that filtered data aligns closely with the original series.
- *Robust to Different Data Frequencies:*
 - ▶ Performs well with quarterly, monthly, or annual data, making it highly versatile.
 - ▶ Recommended horizon h can be adjusted for different data frequencies (e.g., $h = 8$ for quarterly, $h = 24$ for monthly).

Applying MA or Hamilton Filter

- Recall the output decomposition formula:

$$Y_t = \bar{Y}_t + \tilde{Y}_t$$

- The term $\beta_0 + \beta_1 y_t + \beta_2 y_{t-1} + \beta_3 y_{t-2} + \beta_4 y_{t-3}$ can be interpreted as the potential output component ($\bar{Y}_t$)
- The term v_{t+h} corresponds to the cyclical component ($\tilde{Y}_t$)
- To improve statistical properties and simplify analysis, GDP is often transformed by taking its logarithm